

1. Create a Class and Object

Objective: Learn how to define a class and create an object.

```
class Dog:

    def bark(self):

        print("Woof!")
```

Create object

```
my_dog = Dog()

my_dog.bark()
```

2. Class with __init__ Method

Objective: Learn to use the constructor to initialize attributes.

```
class Student:

    def __init__(self, name, grade):

        self.name = name

        self.grade = grade

    def display(self):

        print(f"{self.name} is in grade {self.grade}")
```

Object

```
s1 = Student("Alice", 8)

s1.display()
```

3. Add and Use Class Methods

Objective: Add behavior using methods.

```
class Circle:

    def __init__(self, radius):

        self.radius = radius
```

```
def area(self):  
    return 3.14 * self.radius * self.radius
```

Object

```
c = Circle(5)  
print("Area:", c.area())
```

4. Class with Default Values

Objective: Use default values for parameters.

```
class Lamp:  
    def __init__(self, color="White"):  
        self.color = color  
  
    def switch_on(self):  
        print(f"{self.color} lamp is on.")
```

```
lamp1 = Lamp()  
lamp1.switch_on()
```

```
lamp2 = Lamp("Red")  
lamp2.switch_on()
```

5. Inheritance

Objective: Understand how to inherit from a class.

```
class Animal:
```

```
def speak(self):  
    print("Animal speaks")  
class Cat(Animal):  
    def speak(self):  
        print("Meow")  
a = Animal()  
a.speak()
```

```
c = Cat()  
c.speak()
```

Create a class Car with a method drive() that prints "Vroom!"

Create a class Book with title and author, and a method to display them.

Create a class Rectangle with length and width, and method area().

Create a class Fan with a default speed of "Medium" and a method status().

Create a class Bird that inherits from Animal and overrides speak() to say "Tweet"